Razi University, Kermanshah, Iran
M.Sc. in Artificial Intelligence
Graduate Research Assistant

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Nasim Beigzadeh
GitHub Profile
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Summary

Researcher with **4+ years** of experience at the Signal and Image Processing Laboratory, specializing in **medical imaging, biomedical signal processing, and deep learning for healthcare applications**. Experienced in designing lightweight and computationally efficient deep learning architectures through **knowledge distillation and model compression**, applied to cardiovascular disease diagnosis from ECG signals, with performance suitable for low-power and resource-constrained healthcare devices. Author of **four journal papers extracted from my M.Sc. thesis**.Proficient in Python, PyTorch, TensorFlow, and OpenCV. Seeking Ph.D. research opportunities in **artificial intelligence and deep learning**, with broader interests spanning medical applications, computer vision, and intelligent systems.

Education

M.Sc. in Artificial Intelligence, Computer Engineering *2021–2024*
Razi University, Kermanshah, Iran
- Thesis: *Diagnosis of Cardiac Diseases by Applying a Combination of Knowledge Distillation and Transformer Model on ECG Signals.*
- Supervisor: Prof. Abdolhossein Fathi.
- Thesis Grade: **A**$^+$
- Total CGPA: **3.92/4.00** (WES iGPA)

B.Sc. in Hardware Engineering, Computer Engineering *2014–2020*
University of Science and Technology of Mazandaran, Mazandaran, Iran
- Thesis: *Design and Implementation of a Smart Irrigation System Using Arduino.*
- Thesis Grade: **A**$^+$

Research Interests

- Imaging & Signal Processing
- Lightweight Neural Network Design
- Neuromorphic Spiking Neural Networks
- Image Segmentation
- Machine Learning & Deep Learning
- IoT & AI-driven Intelligent Systems

Technical Skills

Programming Languages: Python, Arduino (basic)
Deep Learning Frameworks: PyTorch, TensorFlow, Keras
Scientific Libraries: NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Seaborn
Computer Vision: OpenCV, image preprocessing, and augmentation techniques

Publications

Journal articles (Published)

- **Beigzadeh, N.**, & Fathi, A. (2026). HMT-KD: Lightweight Deep Learning Model for ECG Analysis Using Hierarchical Multi-Teacher Knowledge Distillation Framework. **Applied Soft Computing** (Q1, IF $\approx$ 7.8).
- **Beigzadeh, N.**, & Fathi, A. (2026). Enhancing Cardiovascular Disease Diagnosis through Knowledge Distillation and Hybrid Transformer–CNN–SENet Architecture. **Journal of Medical Engineering & Technology**.
- **Beigzadeh, N.**, & Fathi, A. (2025). Employing a Deep Learning-based Model on Printed Images of ECG Signals for Cardiovascular Disease Detection. **Journal of Machine Vision and Image Processing**.

Under Review

- **Beigzadeh, N.**, Shahsavari, M., & Fathi, A. (2026). EIA-CKD: An End-to-End Internal Augmentation and Curriculum-Based Knowledge Distillation Framework for Lightweight ECG Classification. **Biomedical Signal Processing and Control** (Q1, IF $\approx$ 5.7).
- Salehnia, T., **Beigzadeh, N.**, & Fathi, A. (2026). Natural Image Segmentation Using Metaheuristic Algorithms.

Research Experience

Research Collaborator *2025–Present*
Donders Institute for Brain, Cognition and Behaviour, Radboud University, The Netherlands
Supervisor: Dr. Mahyar Shahsavari Co-supervisor: Prof. Abdolhossein Fathi

- Developed EIA-CKD, a lightweight curriculum-based knowledge distillation framework for cardiovascular disease classification from ECG images, evaluated on the PTB-XL and Chapman datasets, targeting efficient diagnosis for resource-constrained and wearable healthcare systems.

Graduate Research Assistant *Sep 2023–Jul 2025*
Signal and Image Processing Laboratory (SIPL), Razi University, Kermanshah, Iran
Supervisor: Prof. Abdolhossein Fathi

- Conducted research in medical imaging, biomedical signal processing, and deep learning.
- Developed lightweight deep learning architectures for cardiovascular disease diagnosis using ECG signals.
- Worked extensively on knowledge distillation and model compression for computationally efficient neural networks.
- Designed hybrid CNN–SENet–Transformer architectures for ECG classification.
- Investigated efficient deep learning models suitable for low-power and resource-constrained healthcare devices.

Teaching Experience

Teaching Assistant – Neural Networks *Jan 2025–Jun 2025*
Department of Computer Engineering, Razi University
Instructor: Dr. Abdollah Chalechale

- Assisted in teaching and evaluating coursework related to neural network architectures.
- Held weekly tutorial sessions and guided students on implementation projects using PyTorch, NumPy, and Pandas.

Educational Activities

Instructor – Neural Network Functions in Python & Deep Learning for ECG Signal and Motion Analysis *2025*
Farazan Knowledge-Based Startup Center, Razi University Science and Technology Park
Under the supervision of Dr. Abdollah Chalechale.

Peer Review Experience

- Completed **16 peer reviews** for publications and grants.
- Completed **14 reviews for Knowledge-Based Systems** (Q1).
- Completed **1 review for Applied Soft Computing** (Q1).
- Completed **1 review for Biomedical Signal Processing and Control** (Q1).
- Peer-review activity conducted through ORCID: ORCID Profile.

Selected Technical Projects

- Medical image segmentation using U-Net
- Knowledge distillation for model compression
- ECG image generation using GANs
- Hybrid CNN–Transformer model optimization for ECG
- Fine-tuning ResNet, DenseNet, and related architectures for diagnostic imaging
- RNN/LSTM modeling for temporal ECG classification
- Neural Architecture Search (NAS) for optimized network design

Honors & Achievement

- Ranked 3rd of graduating class among MSc students, Artificial Intelligence, Department of Electrical and Computer Engineering, Razi University *2024*

Languages

English: TOEFL iBT (Nov. 2026) **Persian:** Native **Kurdish:** Native

References

Prof. Abdolhossein Fathi *Full Professor, Department of Computer Engineering and Information Technology, Razi University*
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Dr. Mahyar Shahsavari *Assistant Professor, Donders Institute for Brain, Cognition and Behaviour, Radboud University*
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